

**The National University of Singapore
School of Computing**

CS4226 – Internet Architecture

Tutorial 3

Week 5: September 7 - September 11, 2026

NOTE:

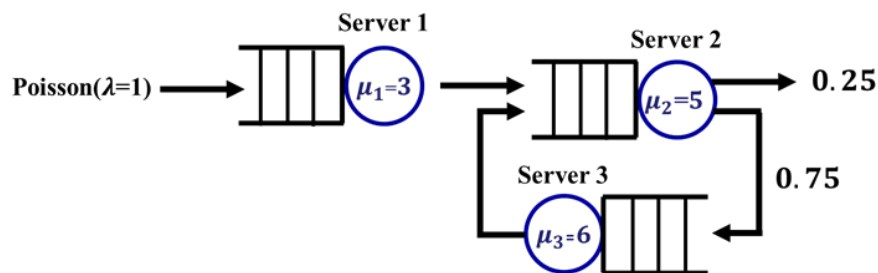
- Please work through the tutorial problems before class and bring your solutions with you.
- If any part of a question is unclear, you are encouraged to ask on Piazza.

Q1. Suppose the arrivals of A2 and D1 buses follow Poisson processes with rate λ_{A2} and λ_{D1} . Suppose you want to go to the central library from SoC and can take either A2 or D1. You will take the A2 or D1 bus that arrives the earliest. What is the probability that you will take an A2 (rather than a D1) bus?

Q2. Suppose we have two Poisson packet flows going through a link, which is modelled as an M/M/1 system. Flow 1 has an arrival rate $\lambda_1 = 5$ packets/sec and flow 2 has an arrival rate $\lambda_2 = 3$ packets/sec. The link has a total capacity of $\mu = 10$ packets/sec. What is the average time for the packets to go through this link? If we separate the link into two channels with capacity μ_1 and μ_2 such that $\mu_1 + \mu_2 = \mu$, and serve flow 1 with capacity μ_1 and serve flow 2 with capacity μ_2 . How can you allocate the capacity for the two flows such that packets in both flows will spend the same amount of time going through the link? What is their average time going through the link now?

Q3. Suppose capacities μ_1 and μ_2 can be merged as $\mu = \mu_1 + \mu_2$. Show that under M/M/1, for any two flows that can be supported by μ_1 and μ_2 , when you merge both and serve them by the aggregated capacity μ , the mean sojourn time is smaller than the minimum of the original mean sojourn times of the two packet flows. Can we derive the same result for queueing delay?

Q4.



The above figure shows a queueing network with three FIFO servers. All servers have exponentially distributed service times with rates as shown. Outside arrivals occur according to a Poisson process with rate $\lambda = 1$ packet/sec. The edges of the network indicate routing probabilities (assume 1 if none shown). What is the average sojourn time of the packets, $E[W]$?